Developing Qt / C++ Projects

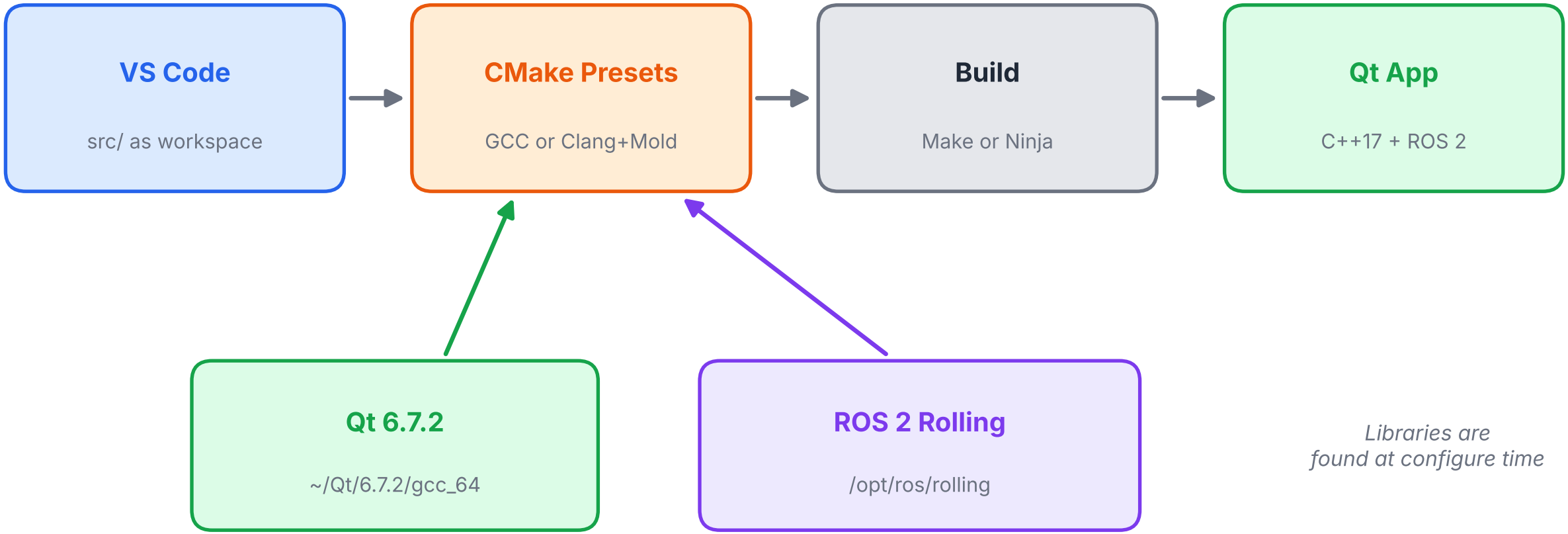
on Ubuntu with VS Code

A simple guide to the daily workflow

Qt 6.7.2 | C++17 | ROS 2 Rolling | CMake Presets

The big picture

Everything is pre-configured. Open, build, debug.



Libraries are found at configure time

What you need installed

Ten tools, five groups

Framework	Build	Compilers	Speed	Dev tools
Qt 6.7.2 ~/Qt/6.7.2/gcc_64	**CMake 3.21+** preset version 6	**Clang 18** default preset	**ccache** compile cache	**GDB** debugger
ROS 2 Rolling /opt/ros/rolling	**Ninja** Clang+Mold backend	**GCC** alternative preset	**mold** fast linker	**VS Code** editor

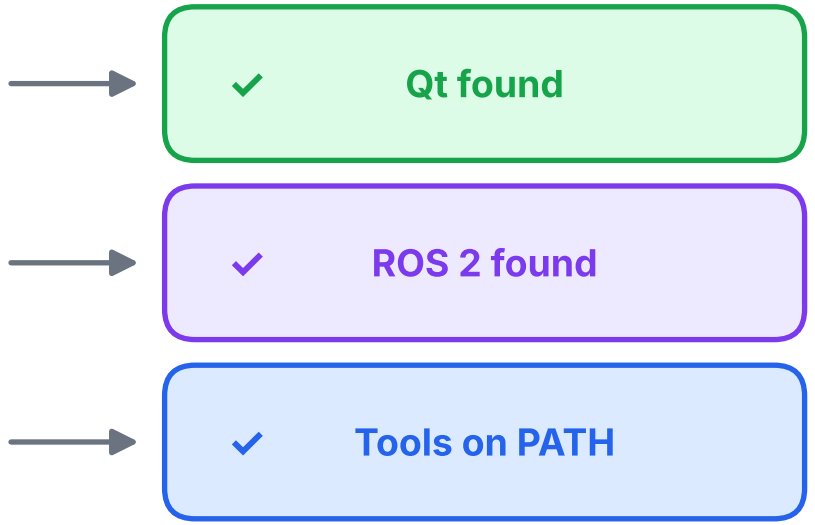
Verify your setup

Run these three checks in a terminal

```
$ ls ~/Qt/6.7.2/gcc_64/lib/cmake/Qt6/Qt6Config.cmake

$ ls /opt/ros/rolling/setup.bash

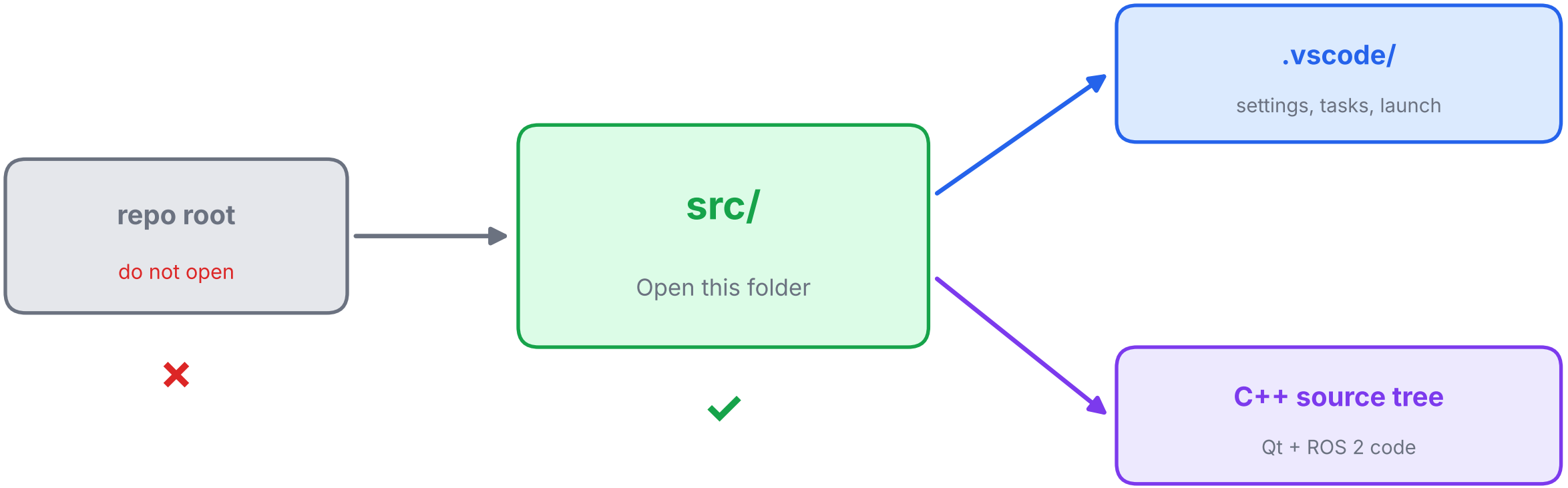
$ which clang++-18 ninja ccache mold gdb
```



All three must pass before you build.

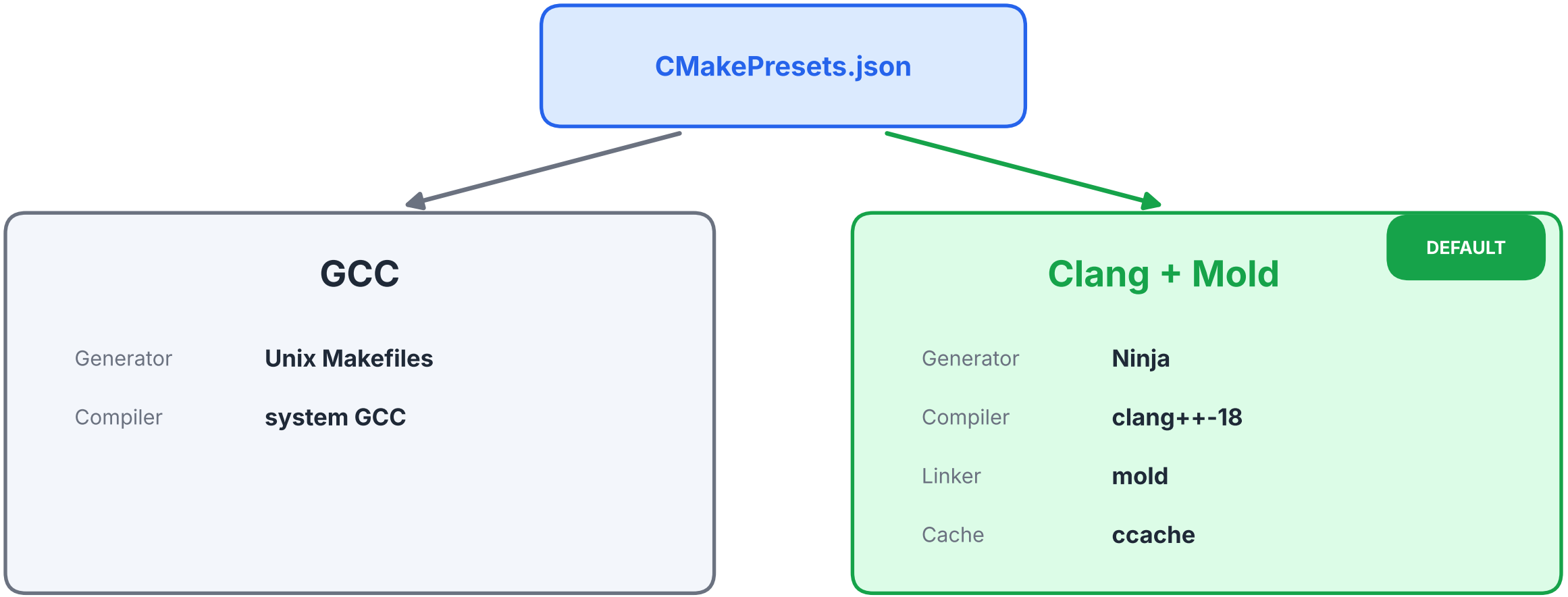
Open the right folder

Open src/ as the workspace, not the repo root



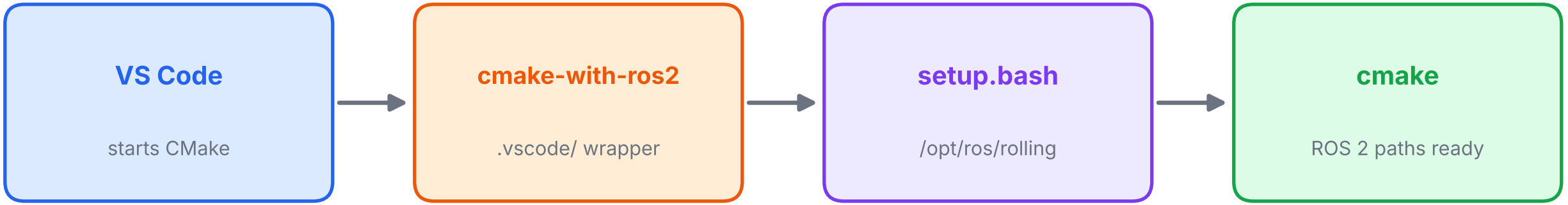
Two build presets

Both are defined in CMakePresets.json



ROS 2 is set up for you

A small wrapper script runs before CMake

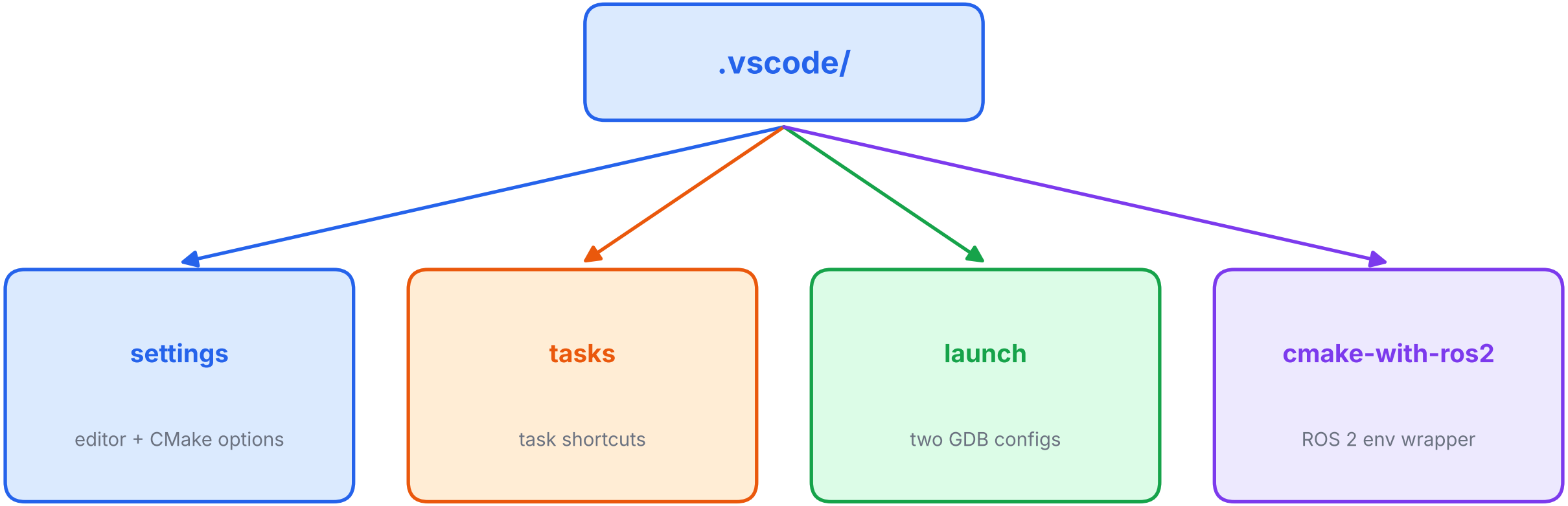


sources the environment

No need to source setup.bash in every terminal

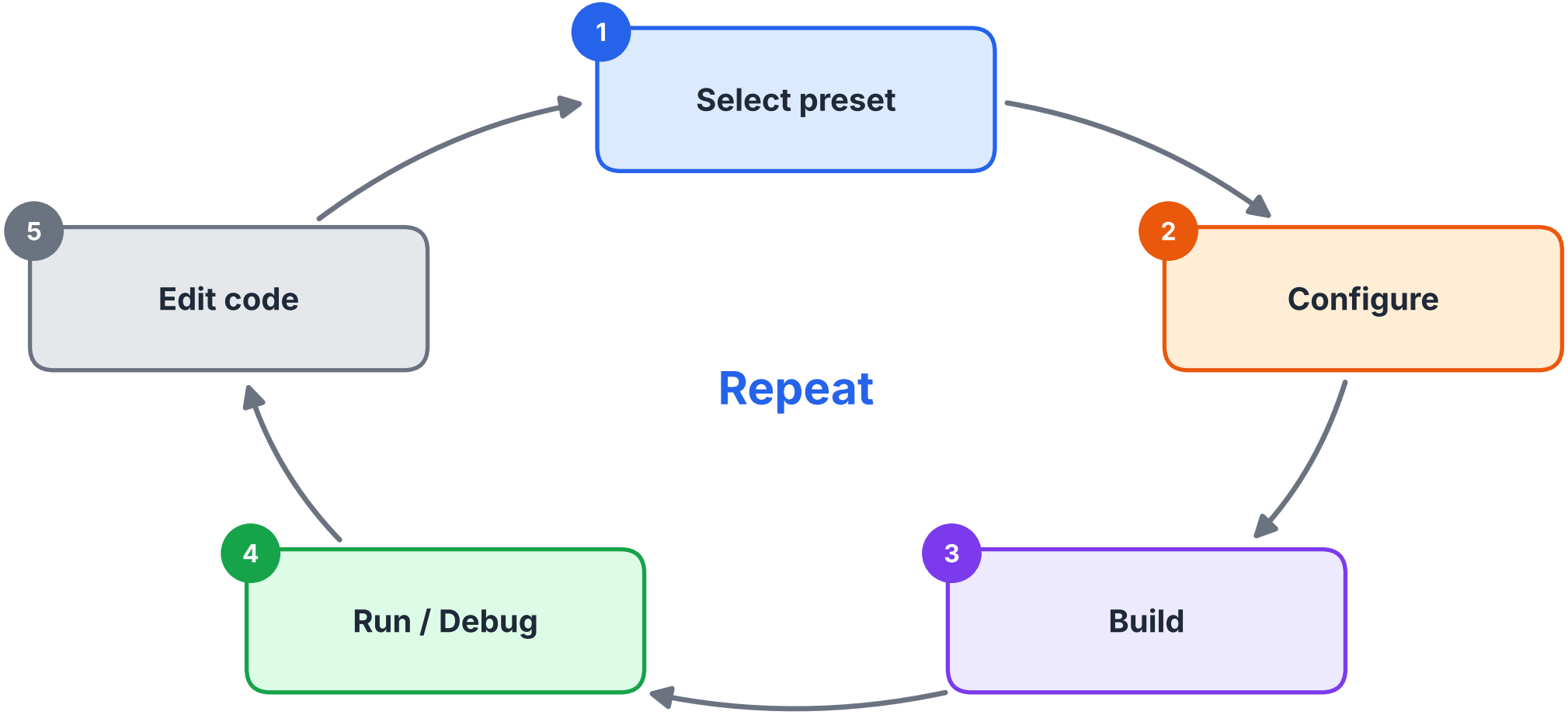
The .vscode/ folder

Pre-configured. Normal work needs no edits.



Your daily loop

Five steps, repeat all day



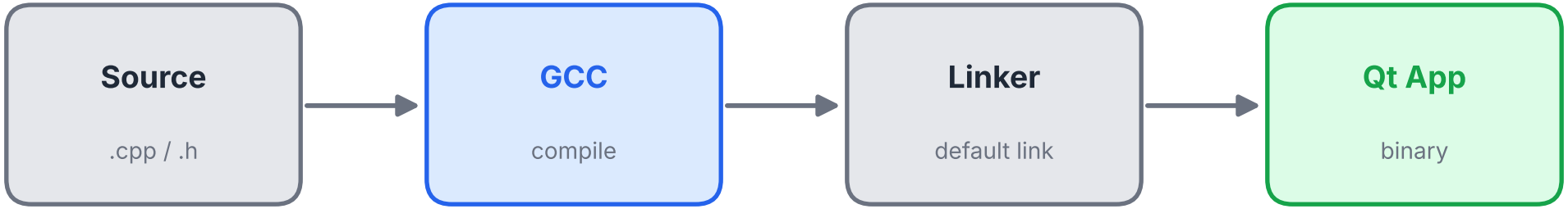
Why the default preset is fast

ccache skips repeat work. mold links quickly.

Default: Clang + Mold



Alternative: GCC



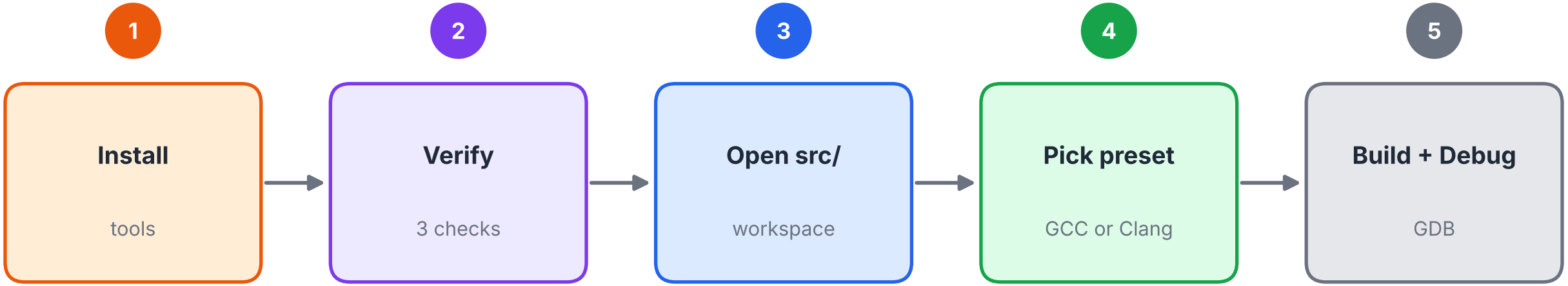
Troubleshooting checklist

Find the symptom, check one thing, fix it

SYMPTOM	CHECK	FIX
Qt not found at configure	Qt6Config.cmake exists?	Install Qt 6.7.2 in ~/Qt
ROS 2 headers missing	/opt/ros/rolling/setup.bash?	Install ROS 2 Rolling
Compiler or tool not found	which clang++-18 ninja ...	Install the missing tool
Presets or tasks missing	Is the workspace src/ ?	Reopen the folder at src/

Summary

From a clean machine to a running Qt app



Everything else is already configured.